

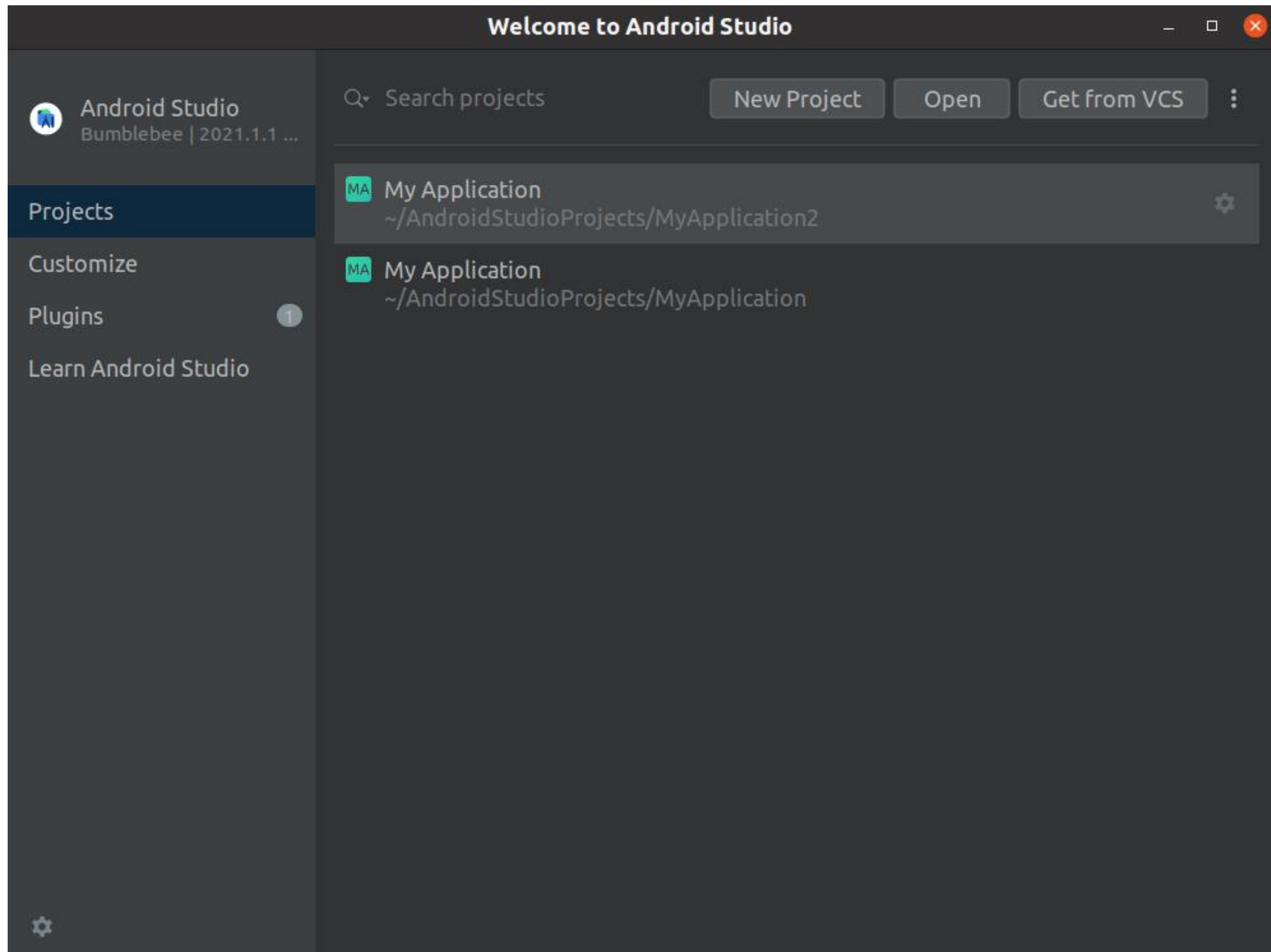
# SEMINARIO DE LENGUAJES OPCIÓN ANDROID + KOTLIN



Primeros pasos con Android

Esp. Fernández Sosa Juan Francisco

Hacer clic en *el menú derecho / SDK Manager*



# Hacer clic en *Configure / SDK Manager*

Appearance & Behavior

Appearance

Menus and Toolbars

System Settings

HTTP Proxy

Data Sharing

Date Formats

Updates

Process Elevation

Passwords

Android SDK

Memory Settings

Notifications

Quick Lists

Path Variables

Keymap

Editor

Build, Execution, Deployment

Languages & Frameworks

Tools

Advanced Settings

Appearance & Behavior > System Settings > Android SDK

Manager for the Android SDK and Tools used by the IDE

Android SDK Location: /Users/jfernandez/Library/Android/sdk

Edit Optimize disk space

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, the IDE will automatically check for updates. Check "show package details" to display individual SDK components.

| Name                                                            | API Level              | Revision | Status              |
|-----------------------------------------------------------------|------------------------|----------|---------------------|
| <input type="checkbox"/> Android UpsideDownCake Preview         | UpsideDownCake         | 1        | Not installed       |
| <input type="checkbox"/> Android TiramisuPrivacySandbox Preview | TiramisuPrivacySandbox | 9        | Not installed       |
| <input type="checkbox"/> Android API 33                         | 33                     | 2        | Partially installed |
| <input type="checkbox"/> Android API 32 (Sv2)                   | 32                     | 1        | Update available    |
| <input checked="" type="checkbox"/> Android 12.0 (S)            | 31                     | 1        | Update available    |
| <input type="checkbox"/> Android 11.0 (R)                       | 30                     | 3        | Partially installed |
| <input checked="" type="checkbox"/> Android 10.0 (Q)            | 29                     | 5        | Installed           |
| <input type="checkbox"/> Android 9.0 (Pie)                      | 28                     | 6        | Partially installed |
| <input type="checkbox"/> Android 8.1 (Oreo)                     | 27                     | 3        | Partially installed |
| <input type="checkbox"/> Android 8.0 (Oreo)                     | 26                     | 2        | Not installed       |
| <input type="checkbox"/> Android 7.1.1 (Nougat)                 | 25                     | 3        | Partially installed |
| <input type="checkbox"/> Android 7.0 (Nougat)                   | 24                     | 2        | Not installed       |
| <input type="checkbox"/> Android 6.0 (Marshmallow)              | 23                     | 3        | Not installed       |
| <input type="checkbox"/> Android 5.1 (Lollipop)                 | 22                     | 2        | Not installed       |
| <input type="checkbox"/> Android 5.0 (Lollipop)                 | 21                     | 2        | Not installed       |

Android 10 instalado

Android 11 parcialmente instalado

Android 12 actualizaciones disponibles

# Hacer clic en *Configure / SDK Manager*

Q

Appearance & Behavior > System Settings > Android SDK

Appearance & Behavior

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Manager for the Android SDK and Tools used by the IDE

Android SDK Location: /Users/jfernandez/Library/Android/sdk Edit Optimize disk space

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, the IDE will automatically check for updates. Check "show package details" to display individual SDK components.

|                                     | Name                                   | API Level              | Revision | Status              |
|-------------------------------------|----------------------------------------|------------------------|----------|---------------------|
| <input type="checkbox"/>            | Android UpsideDownCake Preview         | UpsideDownCake         | 1        | Not installed       |
| <input type="checkbox"/>            | Android TiramisuPrivacySandbox Preview | TiramisuPrivacySandbox | 9        | Not installed       |
| <input type="checkbox"/>            | Android API 33                         | 33                     | 2        | Partially installed |
| <input type="checkbox"/>            | Android API 32 (Sv2)                   | 32                     | 1        | Update available    |
| <input checked="" type="checkbox"/> | Android 12.0 (S)                       | 31                     | 1        | Update available    |
| <input type="checkbox"/>            | Android 11.0 (R)                       | 30                     | 3        | Partially installed |
| <input checked="" type="checkbox"/> | Android 10.0 (Q)                       | 29                     | 5        | Installed           |
| <input type="checkbox"/>            | Android 9.0 (Pie)                      | 28                     | 6        | Partially installed |
| <input type="checkbox"/>            | Android 8.1 (Oreo)                     | 27                     | 3        | Partially installed |
| <input type="checkbox"/>            | Android 8.0 (Oreo)                     | 26                     | 2        | Not installed       |
| <input type="checkbox"/>            | Android 7.1.1 (Nougat)                 | 25                     | 3        | Partially installed |
| <input type="checkbox"/>            | Android 7.0 (Nougat)                   | 24                     | 2        | Not installed       |
|                                     |                                        | 23                     | 3        | Not installed       |
|                                     |                                        | 22                     | 2        | Not installed       |
|                                     |                                        | 21                     | 2        | Not installed       |
|                                     |                                        | 20                     | 2        | Not installed       |
|                                     |                                        | 19                     | 4        | Partially installed |
|                                     |                                        | 18                     | 3        | Not installed       |
|                                     |                                        | 17                     | 2        | Not installed       |

☒ Hide Obsolete Packages ☐ Show Package Details

Desarrollar aplicaciones para versiones antiguas de Android NO requiere instalar las APIs correspondientes a esas versiones



# Hacer clic en *Configure / SDK Manager*

Q

Appearance & Behavior > System Settings > Android SDK

Appearance & Behavior

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Advanced Settings

Manager for the Android SDK and Tools used by the IDE

Android SDK Location:  [Edit](#) [Optimize disk space](#)

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, the IDE will automatically check for updates. Check "show package details" to display individual SDK components.

|                                     | Name                                   | API Level              | Revision | Status              |
|-------------------------------------|----------------------------------------|------------------------|----------|---------------------|
| <input type="checkbox"/>            | Android UpsideDownCake Preview         | UpsideDownCake         | 1        | Not installed       |
| <input type="checkbox"/>            | Android TiramisuPrivacySandbox Preview | TiramisuPrivacySandbox | 9        | Not installed       |
| <input type="checkbox"/>            | Android API 33                         | 33                     | 2        | Partially installed |
| <input type="checkbox"/>            | Android API 32 (Sv2)                   | 32                     | 1        | Update available    |
| <input checked="" type="checkbox"/> | Android 12.0 (S)                       | 31                     | 1        | Update available    |
| <input type="checkbox"/>            | Android 11.0 (R)                       | 30                     | 3        | Partially installed |
| <input checked="" type="checkbox"/> | Android 10.0 (Q)                       | 29                     | 5        | Installed           |
| <input type="checkbox"/>            | Android 9.0 (Pie)                      | 28                     | 6        | Partially installed |
| <input type="checkbox"/>            | Android 8.1 (Oreo)                     | 27                     | 3        | Partially installed |
| <input type="checkbox"/>            | Android 8.0 (Oreo)                     | 26                     | 2        | Not installed       |
| <input type="checkbox"/>            | Android 7.1.1 (Nougat)                 | 25                     | 3        | Partially installed |
| <input type="checkbox"/>            | Android 7.0 (Nougat)                   | 24                     | 2        | Not installed       |
| <input type="checkbox"/>            | Android 6.0 (Marshmallow)              | 23                     | 3        | Not installed       |
| <input type="checkbox"/>            | Android 5.1 (Lollipop)                 | 22                     | 2        | Not installed       |
| <input type="checkbox"/>            | Android 5.0 (Lollipop)                 | 21                     | 2        | Not installed       |
| <input type="checkbox"/>            | Android 4.4W (KitKat Wear)             | 20                     | 2        | Not installed       |
| <input type="checkbox"/>            | Android 4.4 (KitKat)                   | 19                     | 4        | Partially installed |
| <input type="checkbox"/>            | Android 4.3 (Jelly Bean)               | 18                     | 3        | Not installed       |
| <input type="checkbox"/>            | Android 4.2 (Jelly Bean)               | 17                     | 2        | Not installed       |

☒ Hide Obsolete Packages ☐ Show Package Details

Hacer clic aquí

# Hacer clic en *Configure / SDK Manager*

Appearance & Behavior > System Settings > Android SDK

Manager for the Android SDK and Tools used by the IDE

Android SDK Location:  [Edit](#) [Optimize disk space](#)

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, the IDE will automatically check for updates. Check "show package details" to display individual SDK components.

|   | Name                                                                | API Level | Revi... | Status           |
|---|---------------------------------------------------------------------|-----------|---------|------------------|
| > | <input type="checkbox"/> <b>Android API 33, Extension Level 4</b>   |           |         |                  |
| > | <input type="checkbox"/> <b>Android API 32 (Sv2)</b>                |           |         |                  |
| ▼ | <input checked="" type="checkbox"/> <b>Android 12.0 (S)</b>         |           |         |                  |
|   | <input checked="" type="checkbox"/> Android SDK Platform 31         | 31        | 1       | Installed        |
|   | <input checked="" type="checkbox"/> Sources for Android 31          | 31        | 1       | Installed        |
|   | <input type="checkbox"/> Android TV ARM 64 v8a System Image         | 31        | 4       | Not installed    |
|   | <input type="checkbox"/> Android TV Intel x86 Atom System Image     | 31        | 4       | Not installed    |
|   | <input type="checkbox"/> ARM 64 v8a System Image                    | 31        | 4       | Not installed    |
|   | <input type="checkbox"/> Intel x86_64 Atom System Image             | 31        | 4       | Not installed    |
|   | <input type="checkbox"/> Google TV ARM 64 v8a System Image          | 31        | 4       | Not installed    |
|   | <input type="checkbox"/> Google TV Intel x86 Atom System Image      | 31        | 4       | Not installed    |
|   | <input type="checkbox"/> Google APIs ARM 64 v8a System Image        | 31        | 8       | Update Availa... |
|   | <input type="checkbox"/> Google APIs Intel x86_64 Atom System Image | 31        | 12      | Not installed    |
|   | <input type="checkbox"/> Google Play ARM 64 v8a System Image        | 31        | 9       | Not installed    |
|   | <input type="checkbox"/> Google Play Intel x86_64 Atom System Image | 31        | 9       | Not installed    |
| ▼ | <input checked="" type="checkbox"/> <b>Android S Preview</b>        |           |         |                  |
|   |                                                                     |           | 2       | Installed        |
|   |                                                                     |           | 2       | Installed        |

☒ Show Package Details

Apply OK

Se recomienda al menos instalar la plataforma y las fuentes del nivel API con el que se compilarán las aplicaciones



# Hacer clic en *Configure / SDK Manager*

Appearance & Behavior > System Settings > Android SDK

Manager for the Android SDK and Tools used by the IDE

Android SDK Location:  [Edit](#) [Optimize disk space](#)

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the an API level by default. Once installed, the IDE will "show package details" to display individual SDK

**Android 12.0 (S)**

|                                     |                                            |    |    |                  |
|-------------------------------------|--------------------------------------------|----|----|------------------|
| <input checked="" type="checkbox"/> | Android SDK Platform 31                    |    |    |                  |
| <input checked="" type="checkbox"/> | Sources for Android 31                     | 31 | 1  | Installed        |
| <input type="checkbox"/>            | Android TV ARM 64 v8a System Image         | 31 | 4  | Not installed    |
| <input type="checkbox"/>            | Android TV Intel x86 Atom System Image     | 31 | 4  | Not installed    |
| <input type="checkbox"/>            | ARM 64 v8a System Image                    | 31 | 4  | Not installed    |
| <input type="checkbox"/>            | Intel x86_64 Atom System Image             | 31 | 4  | Not installed    |
| <input type="checkbox"/>            | Google TV ARM 64 v8a System Image          | 31 | 4  | Not installed    |
| <input type="checkbox"/>            | Google TV Intel x86 Atom System Image      | 31 | 4  | Not installed    |
| <input type="checkbox"/>            | Google APIs ARM 64 v8a System Image        | 31 | 8  | Update Availa... |
| <input type="checkbox"/>            | Google APIs Intel x86_64 Atom System Image | 31 | 12 | Not installed    |
| <input type="checkbox"/>            | Google Play ARM 64 v8a System Image        | 31 | 9  | Not installed    |
| <input type="checkbox"/>            | Google Play Intel x86_64 Atom System Image | 31 | 9  | Not installed    |
| <input checked="" type="checkbox"/> | <b>Android S Preview</b>                   |    |    |                  |
| <input checked="" type="checkbox"/> | Google Play ARM 64 v8a System Image        | S  | 2  | Installed        |
| <input type="checkbox"/>            | <b>Android 11.0 (R)</b>                    |    |    |                  |
| <input type="checkbox"/>            | Android SDK Platform 30                    | 30 | 2  | Installed        |

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Project-level settings will be applied to new projects

Cancel Apply OK

Opcionalmente puede instalar Google APIs para acceder a los servicios Google (map, sing-in, places, etc.)

# Hacer clic en *Configure / SDK Manager*

Appearance & Behavior > System Settings > Android SDK

Manager for the Android SDK and Tools used by the IDE

Android SDK Location:  [Edit](#) [Optimize disk space](#)

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package contains an API level by default. Once you select a package, click "show package details" to display more information.

Todas estas imágenes pertenecen a dispositivos que podemos emular para correr y depurar las aplicaciones que desarrollemos

| Name                                                                    | API Level | System Image | Size             | Status |
|-------------------------------------------------------------------------|-----------|--------------|------------------|--------|
| > <input type="checkbox"/> Android API 33, I                            |           |              |                  |        |
| > <input type="checkbox"/> Android API 32 (S                            |           |              |                  |        |
| ▼ <input checked="" type="checkbox"/> <b>Android 12.0 (S)</b>           |           |              |                  |        |
| <input checked="" type="checkbox"/> Android SDK Platform 31             | 31        | 1            | Installed        |        |
| <input checked="" type="checkbox"/> Sources for Android 31              | 31        | 1            | Installed        |        |
| <input type="checkbox"/> Android TV ARM 64 v8a System Image             | 31        | 4            | Not installed    |        |
| <input type="checkbox"/> Android TV Intel x86 Atom System Image         | 31        | 4            | Not installed    |        |
| <input type="checkbox"/> ARM 64 v8a System Image                        | 31        | 4            | Not installed    |        |
| <input type="checkbox"/> Intel x86_64 Atom System Image                 | 31        | 4            | Not installed    |        |
| <input type="checkbox"/> Google TV ARM 64 v8a System Image              | 31        | 4            | Not installed    |        |
| <input type="checkbox"/> Google TV Intel x86 Atom System Image          | 31        | 4            | Not installed    |        |
| <input checked="" type="checkbox"/> Google APIs ARM 64 v8a System Image | 31        | 8            | Update Availa... |        |
| <input type="checkbox"/> Google APIs Intel x86_64 Atom System Image     | 31        | 12           | Not installed    |        |
| <input type="checkbox"/> Google Play ARM 64 v8a System Image            | 31        | 9            | Not installed    |        |
| <input type="checkbox"/> Google Play intel x86_64 Atom System image     | 31        | 9            | Not installed    |        |
| ▼ <input checked="" type="checkbox"/> <b>Android S Preview</b>          |           |              |                  |        |
| <input checked="" type="checkbox"/> Google Play ARM 64 v8a System Image | S         | 2            | Installed        |        |
| ▼ <input checked="" type="checkbox"/> <b>Android 11.0 (R)</b>           |           |              |                  |        |
| <input checked="" type="checkbox"/> Android SDK Platform 30             | 30        | 2            | Installed        |        |

☒ Hide Obsolete Packages ☒ Show Package Details

? Project-level settings will be applied to new projects

[Cancel](#) [Apply](#) [OK](#)



Settings

Appearance & Behavior

Appearance

Plugins and Toolbars

System Settings

Access Tokens

HTTP Proxy

Updates

Usage Statistics

Android SDK

Notifications

Workspaces

Variables

Execution, Deployment

Appearance & Behavior > System Settings > Android SDK

Manager for the Android SDK and Tools used by Android Studio

Android SDK Location: C:\Users\leona\AppData\Local\Android\Sdk

SDK Platforms

SDK Tools

SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, Android Studio will automatically check for updates. Check "show package details" to display individual SDK components.

|                                     | Name                                       | API Level | Revision | Status        |
|-------------------------------------|--------------------------------------------|-----------|----------|---------------|
| <input type="checkbox"/>            | Google APIs Intel x86 Atom System Image    | 25        | 2        | Not installed |
| <input type="checkbox"/>            | Google APIs Intel x86 Atom_64 System Image | 25        | 2        | Not installed |
| ▼ <input type="checkbox"/>          | <b>Android 7.0 (Nougat)</b>                |           |          |               |
| <input checked="" type="checkbox"/> | Google APIs                                | 24        | 1        | Installed     |
| <input checked="" type="checkbox"/> | Android SDK Platform 24                    | 24        | 2        | Installed     |
| <input checked="" type="checkbox"/> | Sources for Android 24                     | 24        | 1        | Installed     |
| <input type="checkbox"/>            | Android TV Intel x86 Atom System Image     | 24        | 7        | Not installed |
| <input type="checkbox"/>            | Android Wear ARM EABI v7a System Image     | 24        | 2        | Not installed |
| <input type="checkbox"/>            | Android Wear Intel x86 Atom System Image   | 24        | 2        | Not installed |
| <input type="checkbox"/>            | ARM 64 v8a System Image                    | 24        | 7        | Not installed |
| <input type="checkbox"/>            | ARM EABI v7a System Image                  | 24        | 7        | Not installed |
| <input type="checkbox"/>            | Intel x86 Atom System Image                | 24        | 7        | Not installed |
| <input type="checkbox"/>            | Intel x86 Atom_64 System Image             | 24        | 7        | Not installed |
| <input type="checkbox"/>            | Google APIs ARM 64 v8a System Image        | 24        | 8        | Not installed |
| <input type="checkbox"/>            | Google APIs ARM EABI v7a System Image      | 24        | 8        | Not installed |
| <input checked="" type="checkbox"/> | Google APIs Intel x86 Atom System Image    | 24        | 8        | Installed     |
| <input type="checkbox"/>            | Google APIs Intel x86 Atom_64 System Image | 24        | 8        | Not installed |
| ▼ <input type="checkbox"/>          | <b>Android 6.0 (Marshmallow)</b>           |           |          |               |
| <input type="checkbox"/>            | Google APIs                                | 23        | 1        | Not installed |
| <input checked="" type="checkbox"/> | Android SDK Platform 23                    | 23        | 3        | Installed     |
|                                     |                                            |           | 1        | Not installed |
|                                     |                                            |           | 3        | Not installed |

☒ Show Package Details

OK Cancel Apply Help

Es conveniente emular dispositivos x86 para aprovechar el acelerador Haxm

## Appearance &amp; Behavior

Appearance

Menus and Toolbars

System Settings

Passwords

HTTP Proxy

Updates

Usage Statistics

Android

Icons

Keyboard

Mouse

Voice

Windows

Workspaces

Keyboard

Mouse

Voice

Windows

Workspaces

Keyboard

Mouse

Voice

Windows

Workspaces

## Appearance &amp; Behavior &gt; System Settings &gt; Android SDK

Manager for the Android SDK and Tools used by Android Studio

Android SDK Location: C:\Users\leona\AppData\Local\Android\Sdk

[Edit](#)

## SDK Platforms

## SDK Tools

## SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, Android Studio will automatically check for updates. Check "show package details" to display individual SDK components.

|                                     | Name                                       | API Level | Revision | Status        |
|-------------------------------------|--------------------------------------------|-----------|----------|---------------|
| <input type="checkbox"/>            | Google APIs Intel x86 Atom System Image    | 25        | 2        | Not installed |
| <input type="checkbox"/>            | Google APIs Intel x86 Atom_64 System Image | 25        | 2        | Not installed |
| ▼ <input type="checkbox"/>          | <b>Android 7.0 (Nougat)</b>                |           |          |               |
|                                     |                                            |           | 1        | Installed     |
|                                     |                                            |           | 2        | Installed     |
|                                     |                                            |           | 1        | Installed     |
|                                     |                                            |           | 7        | Not installed |
|                                     |                                            |           | 2        | Not installed |
|                                     |                                            |           | 2        | Not installed |
|                                     |                                            |           | 7        | Not installed |
|                                     |                                            |           | 7        | Not installed |
|                                     |                                            |           | 7        | Not installed |
|                                     |                                            |           | 7        | Not installed |
|                                     |                                            |           | 8        | Not installed |
|                                     |                                            |           | 8        | Not installed |
|                                     |                                            |           | 8        | Installed     |
|                                     |                                            |           | 8        | Not installed |
| <input type="checkbox"/>            | Google APIs ARM EABI v7a System Image      | 24        |          |               |
| <input checked="" type="checkbox"/> | Google APIs Intel x86 Atom System Image    | 24        |          |               |
| <input type="checkbox"/>            | Google APIs Intel x86 Atom_64 System Image | 24        |          |               |
| ▼ <input type="checkbox"/>          | <b>Android 6.0 (Marshmallow)</b>           |           |          |               |
|                                     | Google APIs                                | 23        |          |               |
| <input checked="" type="checkbox"/> | Android SDK Platform 23                    | 23        |          |               |
| <input type="checkbox"/>            | Sources for Android 23                     | 23        |          |               |
| <input type="checkbox"/>            | Android TV ARM EABI v7a System Image       | 23        |          |               |

Tener instalada la plataforma de la API 23, nos permite correr y depurar las aplicaciones que desarrollamos en un dispositivo con Android 6.0

Presionar en el botón  
Cancel

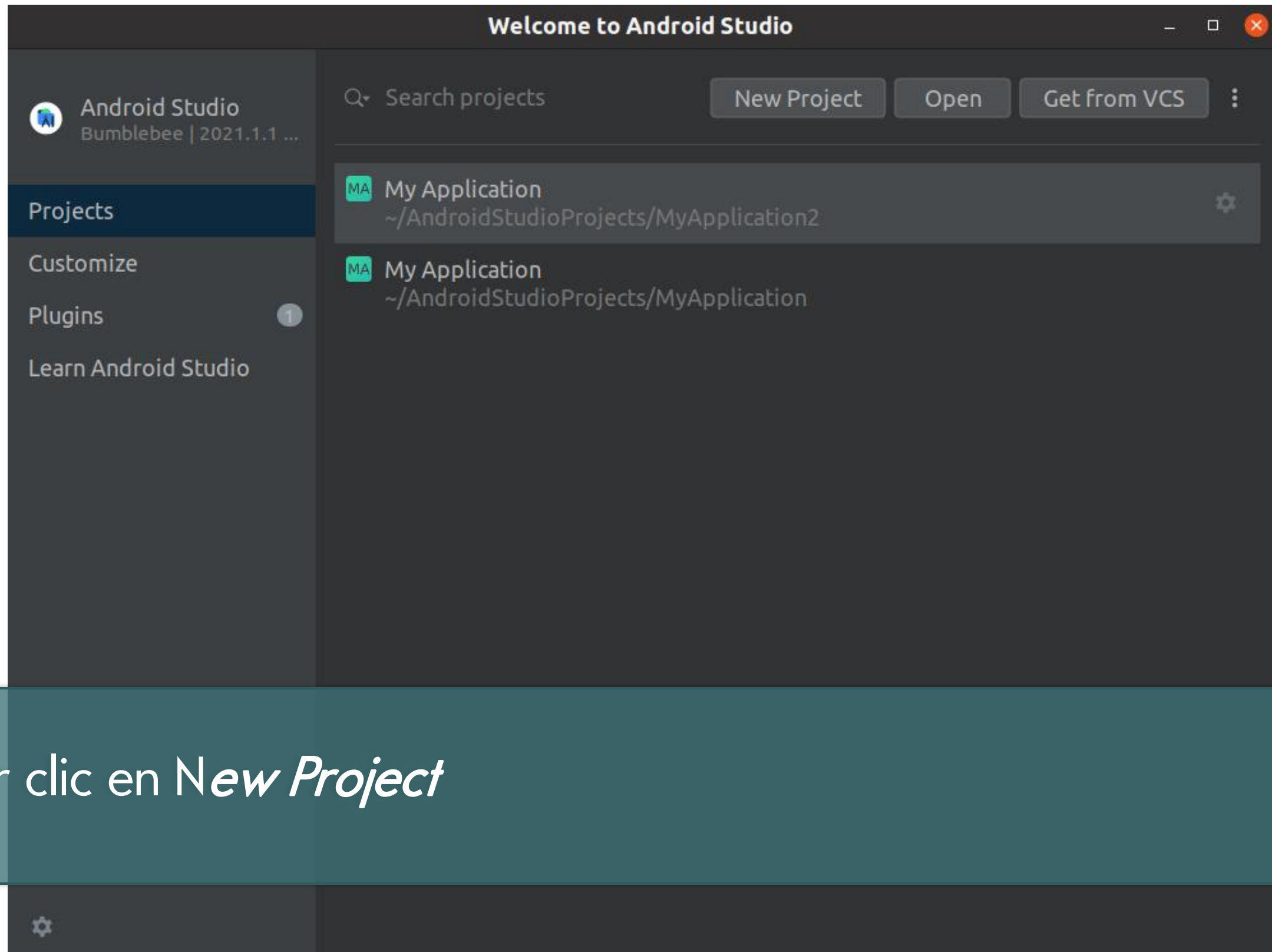
[Launch Standalone SDK Manager](#)

OK

Cancel

Apply

Help



Hacer clic en *New Project*



## Templates

Phone and Tablet

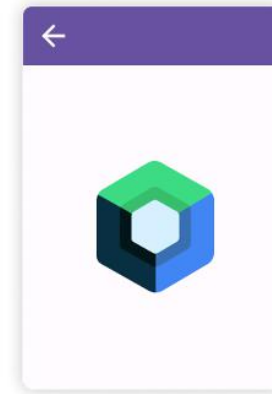
Wear OS

Television

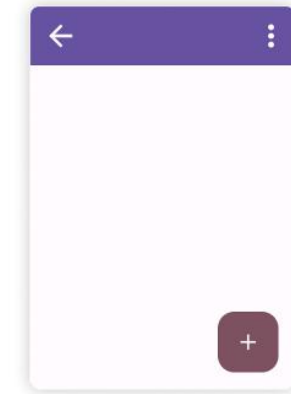
Automotive



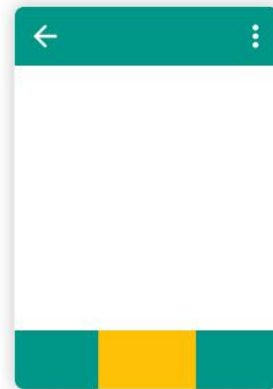
No Activity



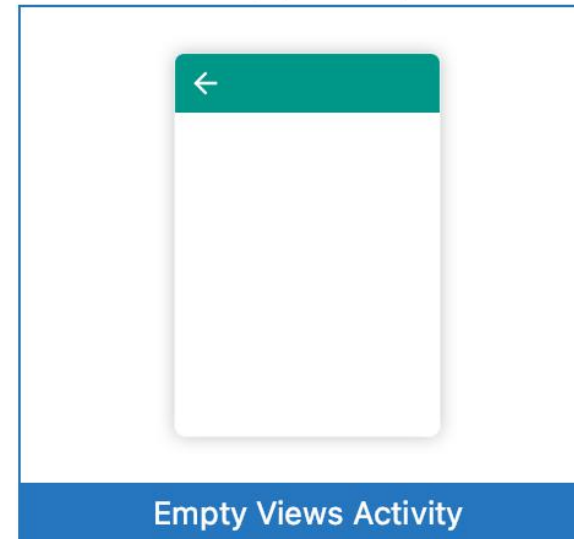
Empty Activity



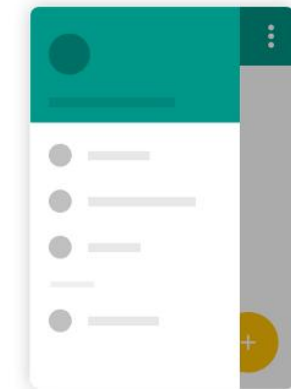
Basic Views Activity



Bottom Navigation Views Activity



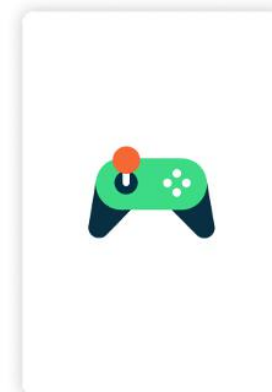
Empty Views Activity



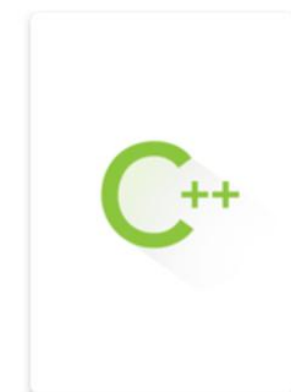
Navigation Drawer Views Activity



Responsive Views Activity



Game Activity (C++)



Native C++

Cancel

Previous

Next

Finish

Debemos elegir Empty Views Activity y luego click en Next

## Empty Views Activity

Creates a new empty activity

|                                                                                                                                                                                           |                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                                                                                                                                                                      | <input type="text" value="My Application"/>                                                                                                                              |
| Package name                                                                                                                                                                              | <input type="text" value="com.example.myapplication"/>                                                                                                                   |
| Save location                                                                                                                                                                             | <input type="text" value="/Users/jfernandez/AndroidStudioProjects/MyApplication10"/>  |
| Language                                                                                                                                                                                  | <input type="text" value="Kotlin"/> ▼                                                                                                                                    |
| Minimum SDK                                                                                                                                                                               | <input 7.0)"="" android="" nougat";="" type="text" value="API 24 ("/> ▼                                                                                                  |
| <p> Your app will run on approximately <b>96.3%</b> of devices.<br/><a href="#">Help me choose</a></p> |                                                                                                                                                                          |
| Build configuration language                                                                           | <input type="text" value="Kotlin DSL (build.gradle.kts) [Recommended]"/> ▼                                                                                               |

Poner un nombre a la aplicación, seleccionar Kotlin como lenguaje y elegir  
Minimum SDK: *API 24*

| ANDROID PLATFORM VERSION |             | API LEVEL | CUMULATIVE DISTRIBUTION |
|--------------------------|-------------|-----------|-------------------------|
| 4.4                      | KitKat      | 19        |                         |
| 5                        | Lollipop    | 21        | 99.6%                   |
| 5.1                      | Lollipop    | 22        | 99.4%                   |
| 6                        | Marshmallow | 23        | 98.2%                   |
| 7                        | Nougat      | 24        | 96.3%                   |
| 7.1                      | Nougat      | 25        | 95.0%                   |
| 8                        | Oreo        | 26        | 93.7%                   |
| 8.1                      | Oreo        | 27        | 91.8%                   |
| 9                        | Pie         | 28        | 86.4%                   |
| 10                       | Q           | 29        | 75.9%                   |
| 11                       | R           | 30        | 59.8%                   |
| 12                       | S           | 31        | 38.2%                   |

## Nougat

### User Interface

Multi-window Support  
Notifications  
Quick Settings Tile API  
Custom Pointer API

### Performance

Profile-guided JIT/AOT Compilation  
Quick Path to App Install  
Sustained Performance API  
Frame Metrics API

### Battery Life

Doze on the Go  
Project Svelte: Background Optimizations  
SurfaceView

### Wireless & Connectivity

Data Saver  
Number Blocking  
Call Screening

### Graphics

Vulkan API

### System

Direct Boot  
Multi-locale Support, More Languages  
ICU4J APIs in Android  
API Signature Scheme?  
Scoped Directory Access  
Keyboard Shortcuts Helper  
Virtual File

### Android for Work

Work profile security challenge  
Turn off work  
Always on VPN  
Customized provisioning

### Accessibility

Vision Settings on the Welcome screen

### Security

Key Attestation  
Network Security Config  
Default Trusted Certificate Authorities

### VR

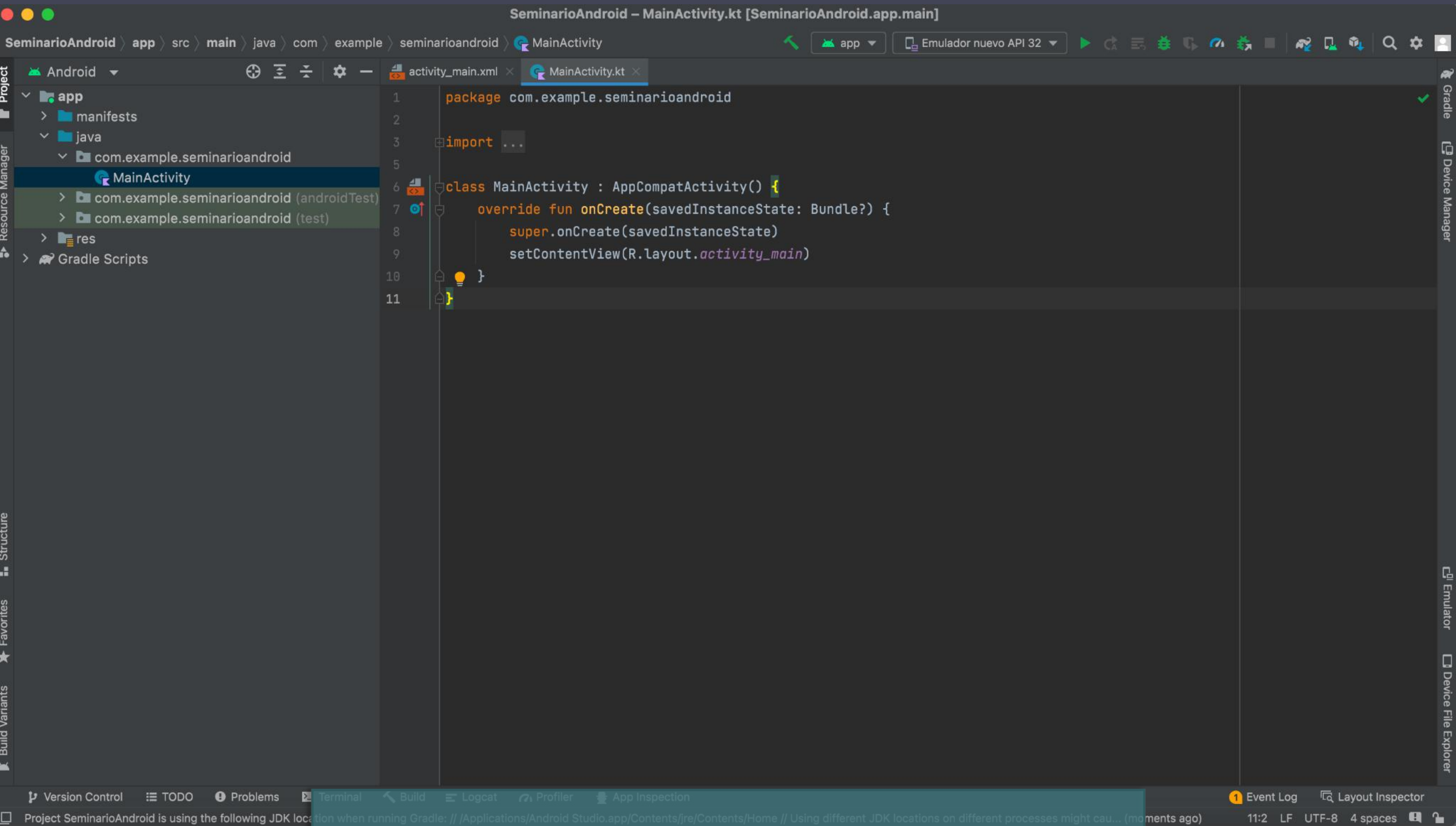
Platform support and optimizations  
VR Mode

### Printing Framework

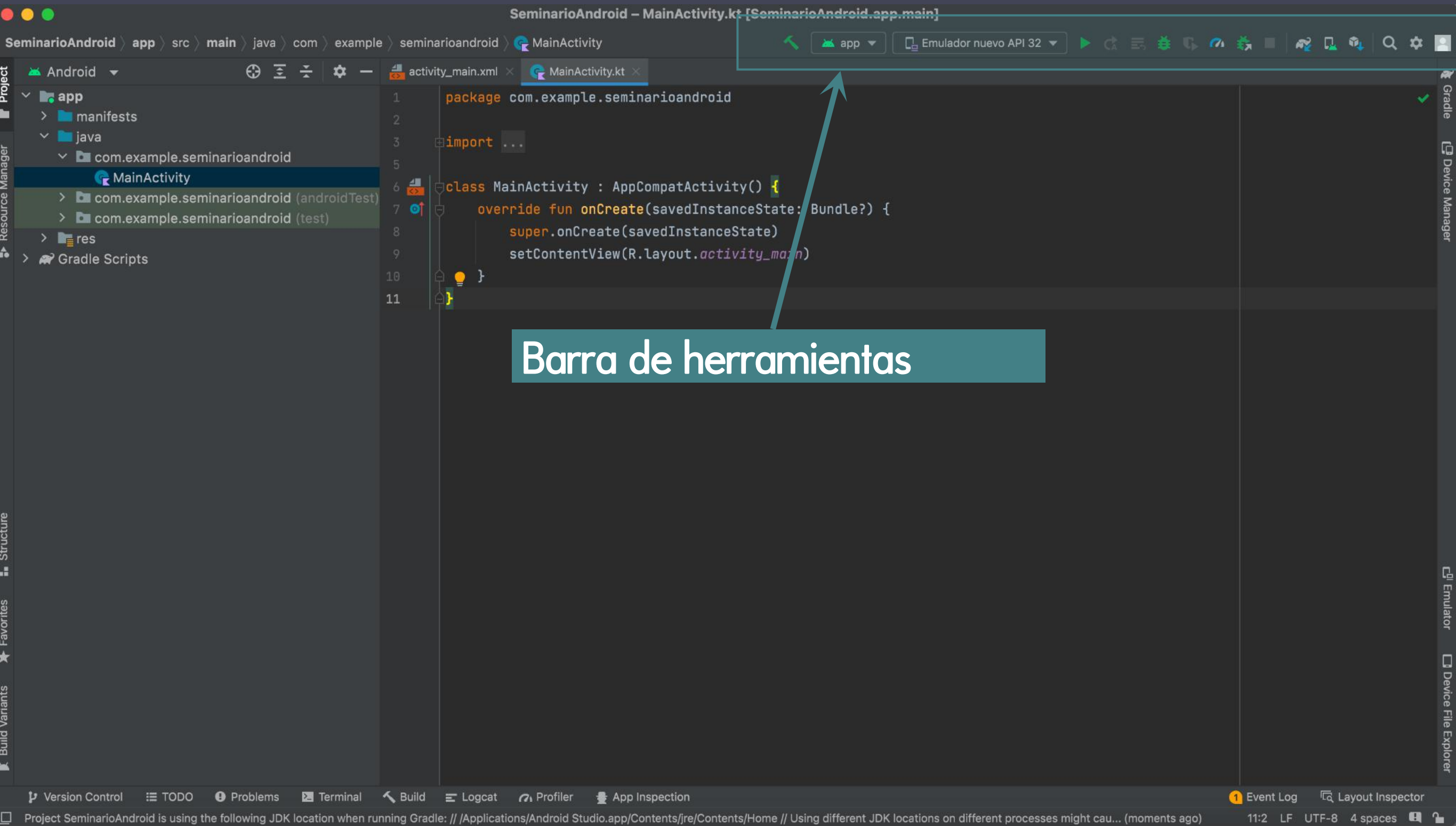
Print service enhancements

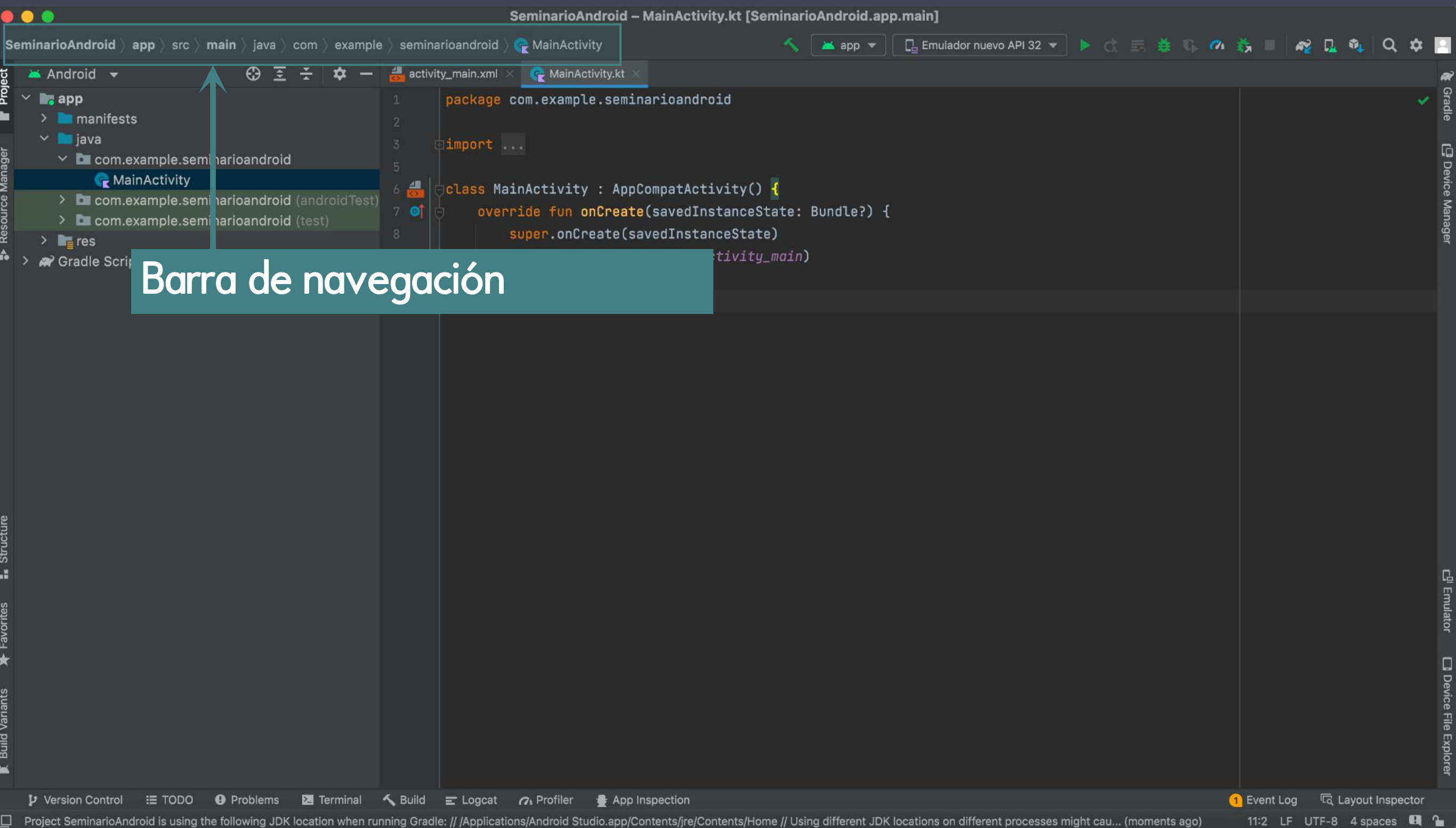
Si se selecciona *help me choose* en la ventana anterior se despliega esta información que puede ayudarnos a elegir la versión mínima soportada por nuestra aplicación





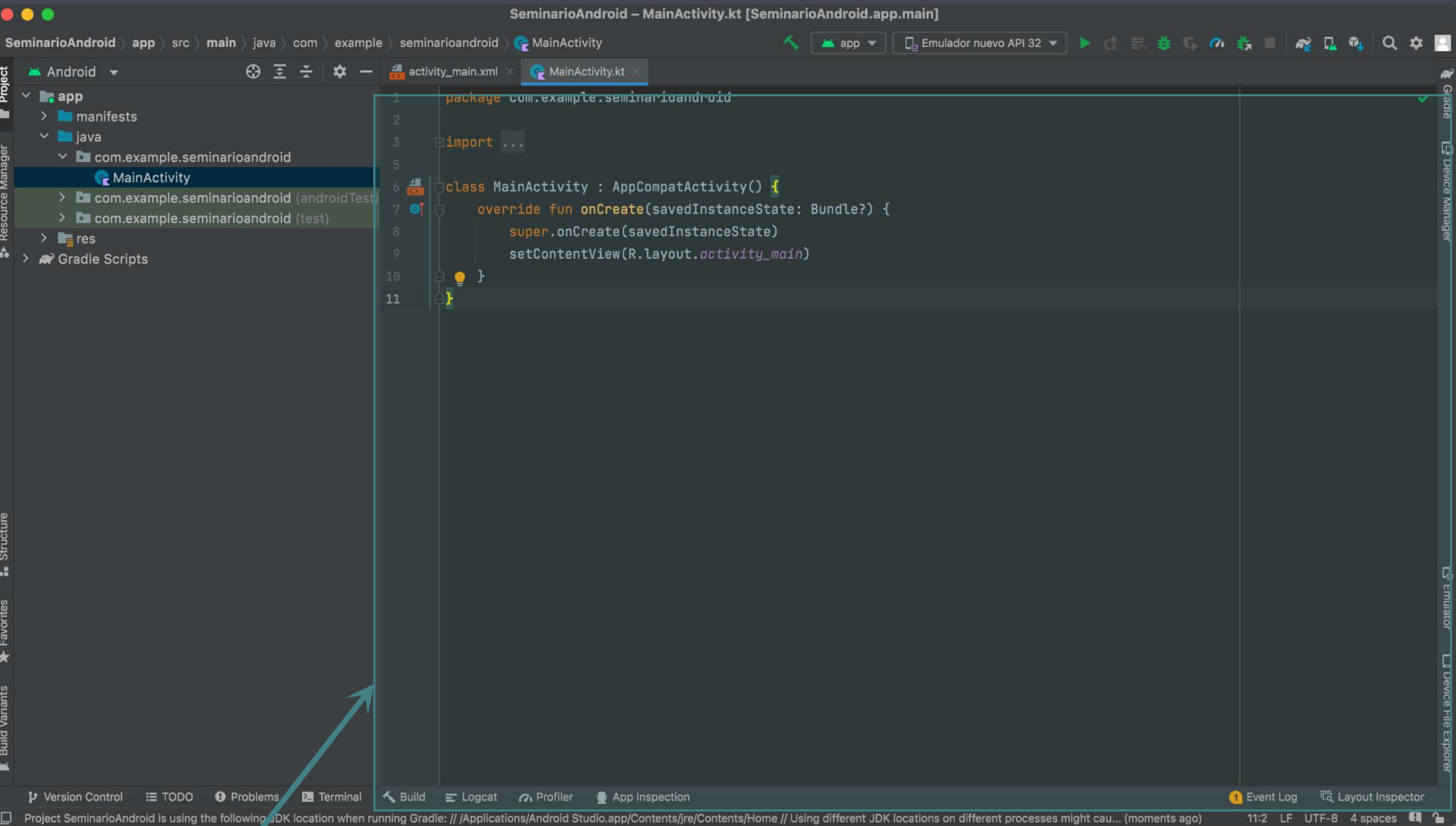
Conociendo la interfaz del IDE



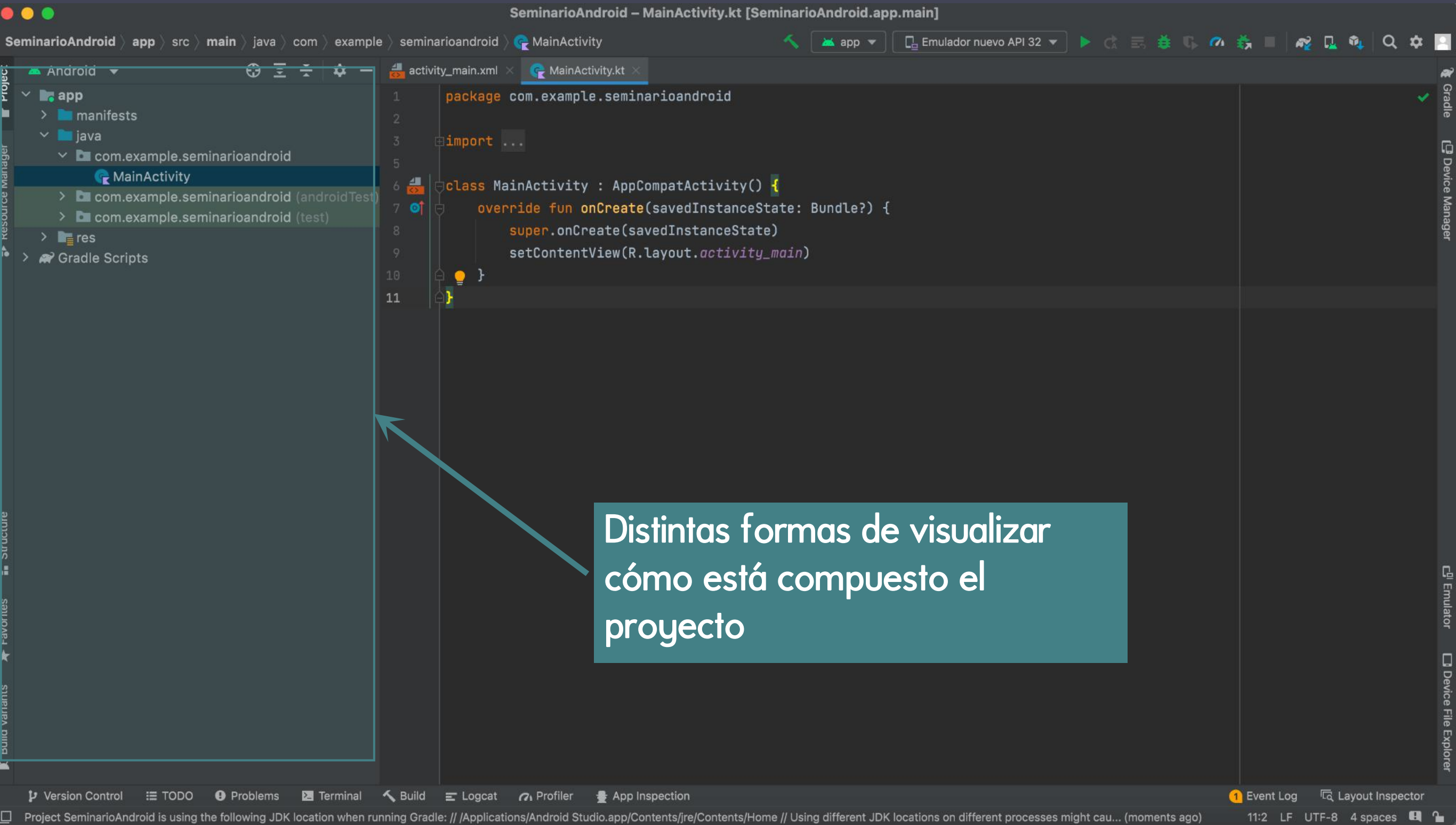


Barra de navegación

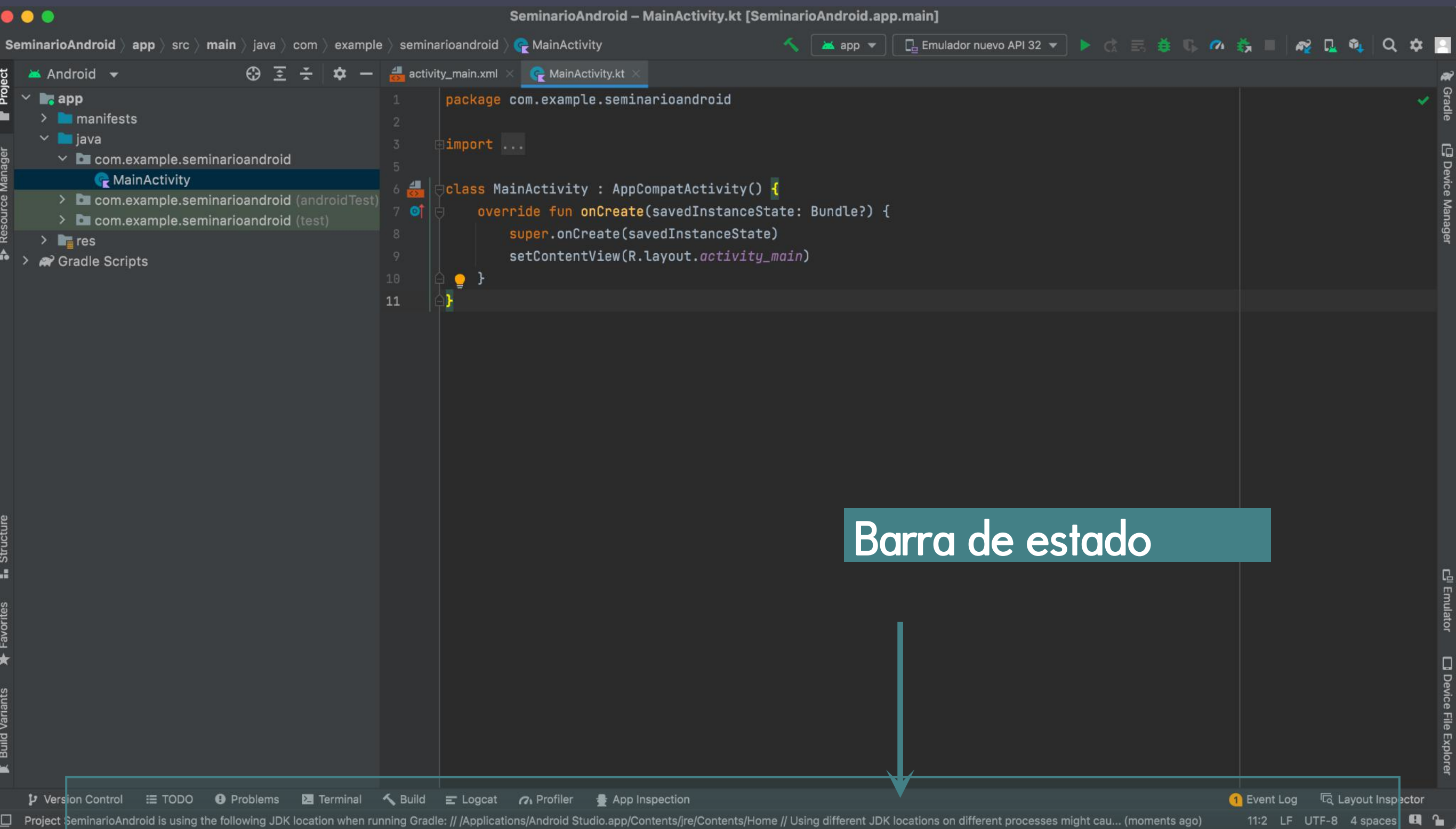




Ventana del editor



Distintas formas de visualizar  
cómo está compuesto el  
proyecto



# Definición de la clase que constituye la Actividad

The screenshot shows the Android Studio interface for a project named "SeminarioAndroid". The Project view on the left shows the package structure: `com.example.seminarioandroid` containing `MainActivity`. The main editor displays the `MainActivity.kt` file with the following code:

```
1 package com.example.seminarioandroid
2
3 import ...
4
5
6 class MainActivity : AppCompatActivity() {
7     override fun onCreate(savedInstanceState: Bundle?) {
8         super.onCreate(savedInstanceState)
9         setContentView(R.layout.activity_main)
10    }
11 }
```

The code defines the `MainActivity` class, which inherits from `AppCompatActivity`. It overrides the `onCreate` method to call `super.onCreate(savedInstanceState)` and `setContentView(R.layout.activity_main)`. A teal box highlights the class definition, and a teal arrow points from the title "Definición de la clase que constituye la Actividad" to the `MainActivity` entry in the Project view.

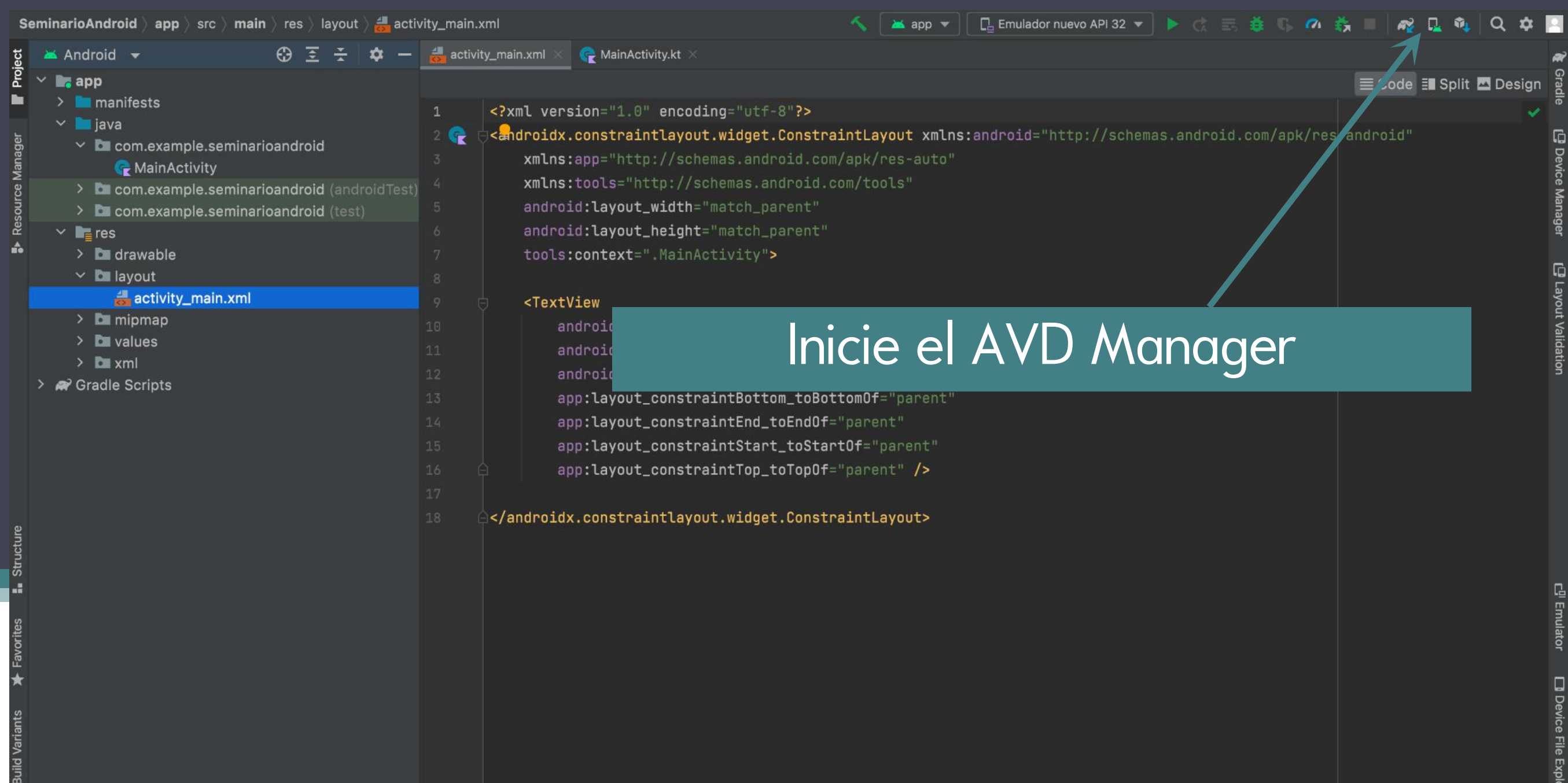


# Definición de la vista de la Actividad

The screenshot displays the Android Studio interface. On the left, the Project view shows the file structure of the 'Seminar' app, with 'activity\_main.xml' selected under the 'res/layout' directory. The main editor window shows the XML code for 'activity\_main.xml'. A teal box highlights the XML code, and a teal arrow points from the title 'Definición de la vista de la Actividad' to this box.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     xmlns:app="http://schemas.android.com/apk/res-auto"
4     xmlns:tools="http://schemas.android.com/tools"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent"
7     tools:context=".MainActivity">
8
9     <TextView
10         android:layout_width="wrap_content"
11         android:layout_height="wrap_content"
12         android:text="Hello World!"
13         app:layout_constraintBottom_toBottomOf="parent"
14         app:layout_constraintEnd_toEndOf="parent"
15         app:layout_constraintStart_toStartOf="parent"
16         app:layout_constraintTop_toTopOf="parent" />
17
18 </androidx.constraintlayout.widget.ConstraintLayout>
```

# Ejecución de la aplicación en el emulador



activity\_main.xml

MainActivity.kt

Code

Split

Design

1

<?xml version="1.0" encoding="utf-8"?>

2

<androidx.constraintlayout.widget.ConstraintLayout xmlns:android

3

xmlns:app="http://schemas.android.com/apk/res-auto"

4

xmlns:tools="http://schemas.android.com/tools"

5

android:layout\_width="match\_parent"

6

android:layout\_height="match\_parent"

14

app:layout\_constraintEnd\_toEndOf="parent"

15

app:layout\_constraintStart\_toStartOf="parent"

16

app:layout\_constraintTop\_toTopOf="parent" />

17

18

</androidx.constraintlayout.widget.ConstraintLayout>

Device Manager

Virtual

Physical

Create device

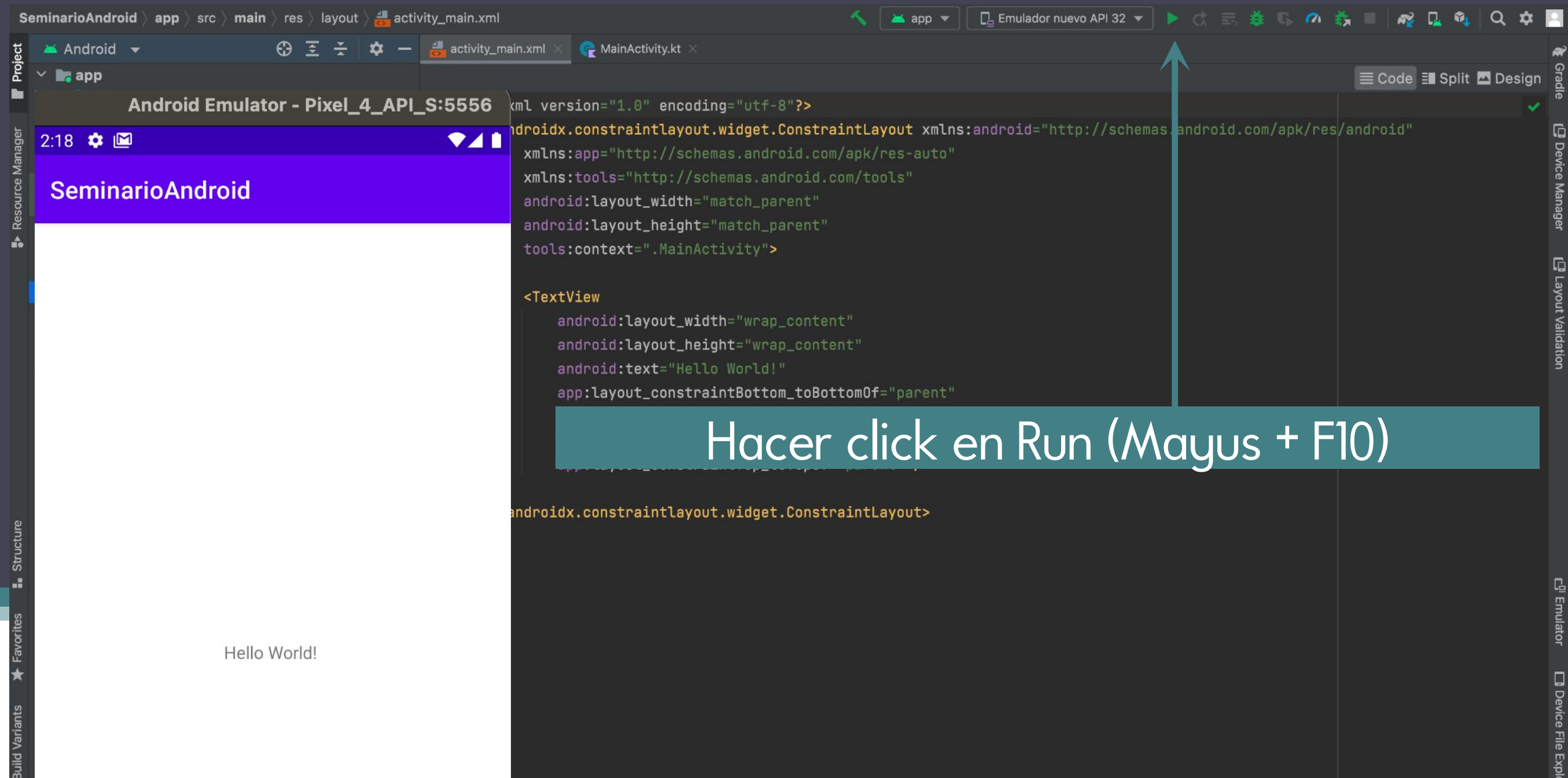
?

| Device                                    | API | Size on Disk | Actions                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|-----|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Emulador nuev...<br>Android API 32 Goo... | 32  | 28 GB        |     |
| Pixel 2 API S<br>Android 12.0 Google...   | 30  | 9.6 GB       |     |
| Pixel 3 API 31<br>Android 12.0 Google...  | 31  | 0.0 MB       |     |
| Pixel 4 API S<br>Android 12.0 Google...   | 30  | 8.0 GB       |     |

Crear un nuevo dispositivo virtual. Ej Pixel 2, con un SDK previamente instalado.

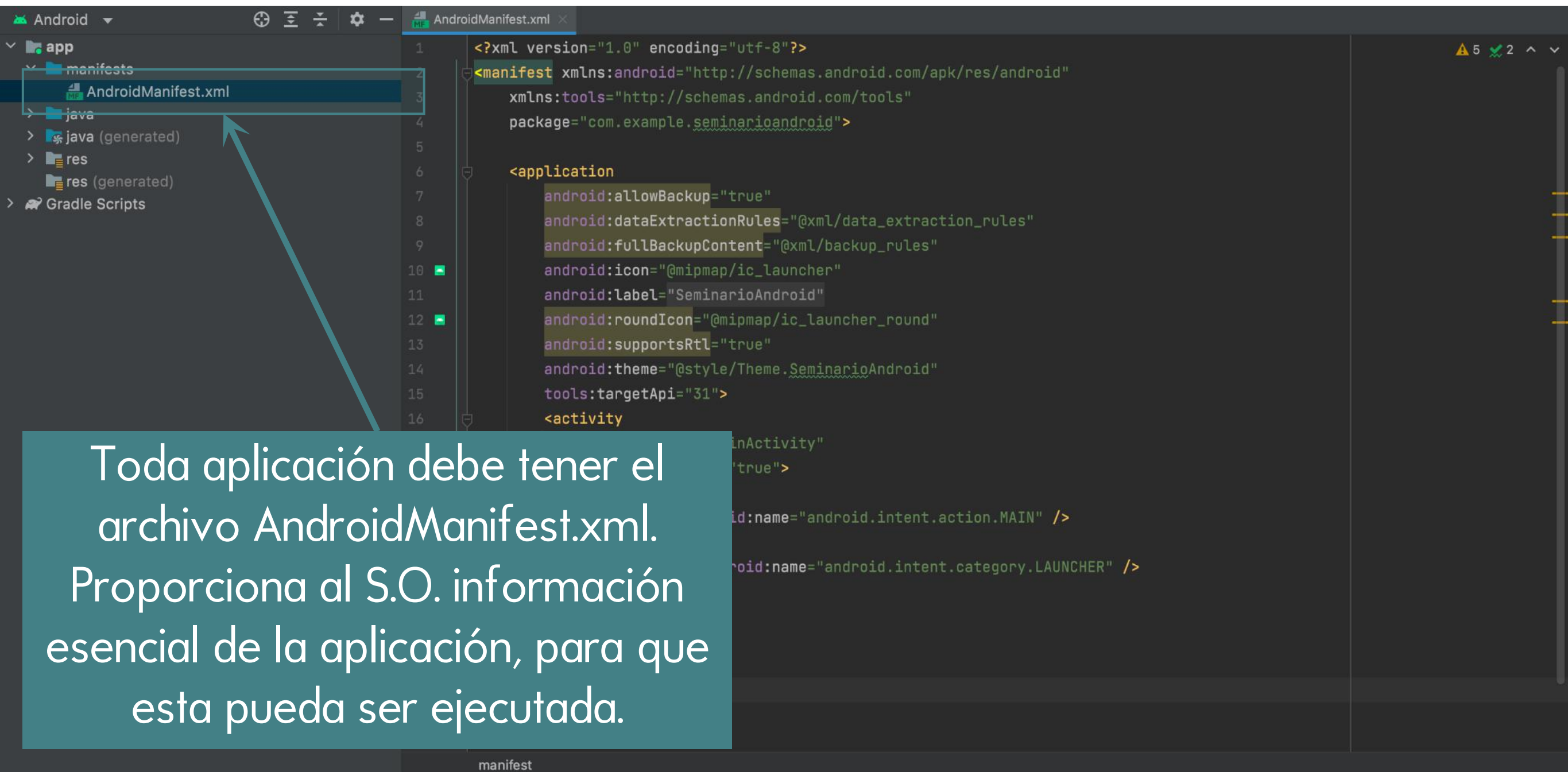
Listado de Emuladores/Dispositivos Virtuales instalados



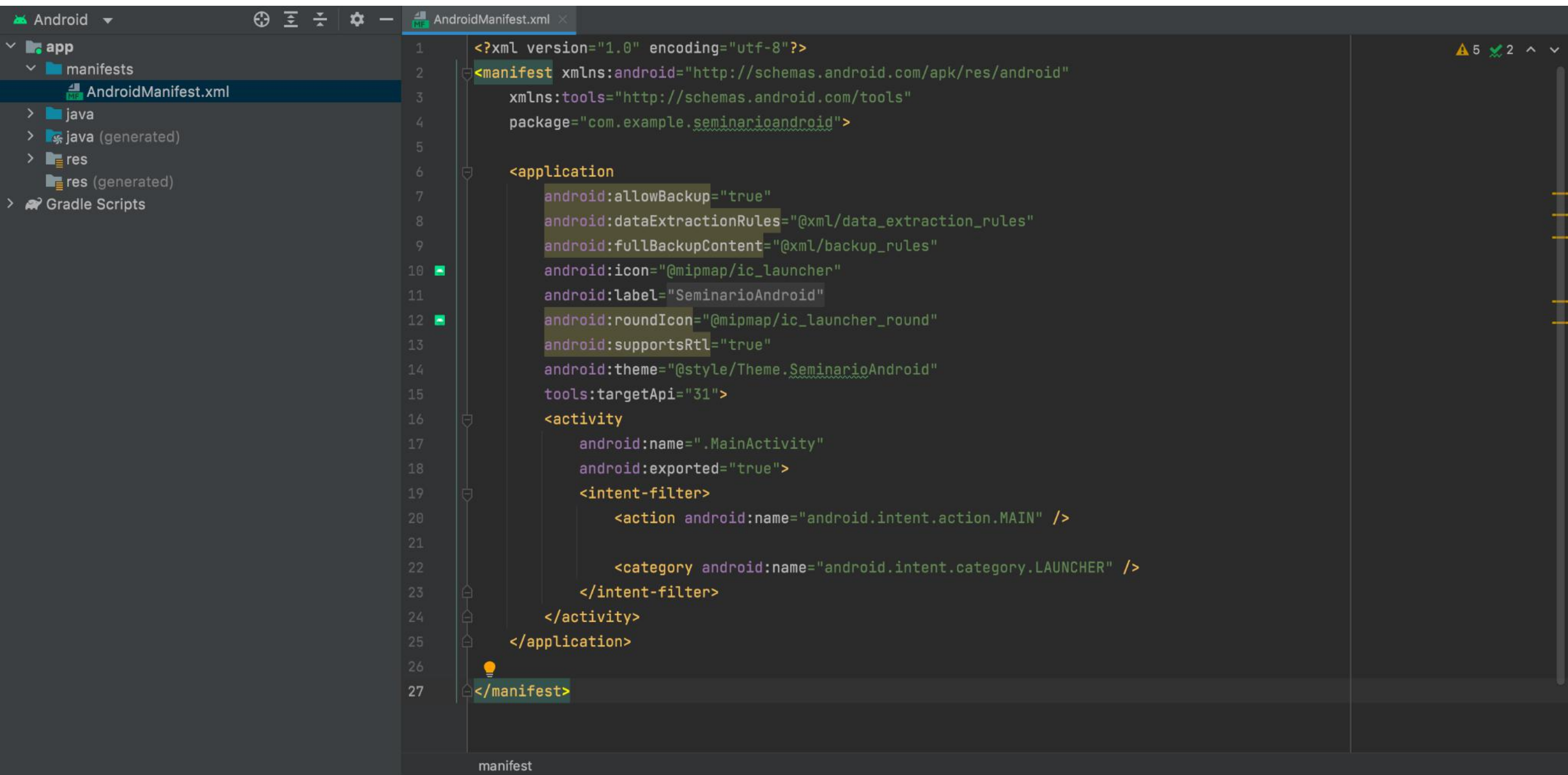


Hacer click en Run (Mayus + F10)

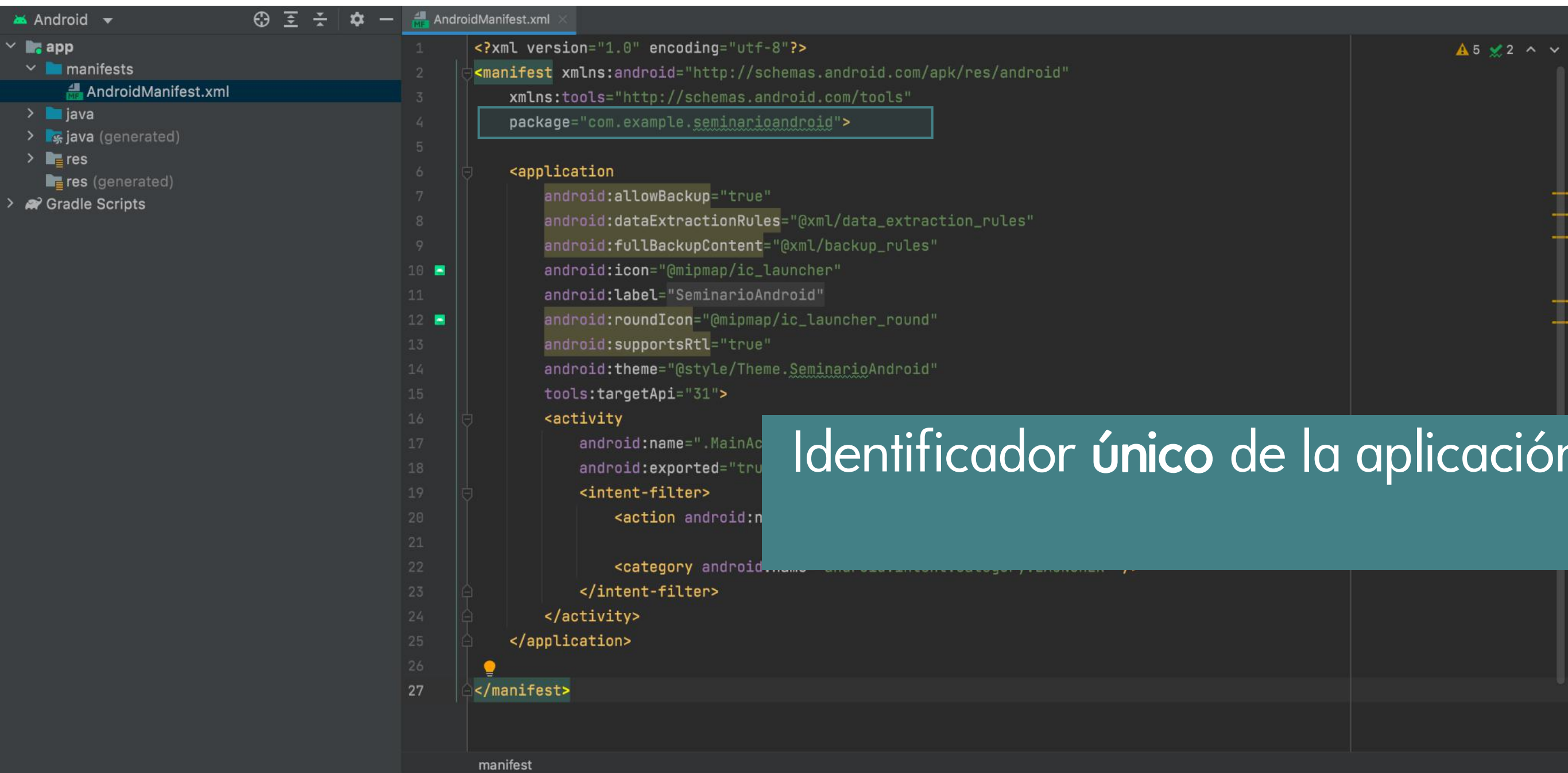
# AndroidManifest.xml



# AndroidManifest.xml



# AndroidManifest.xml





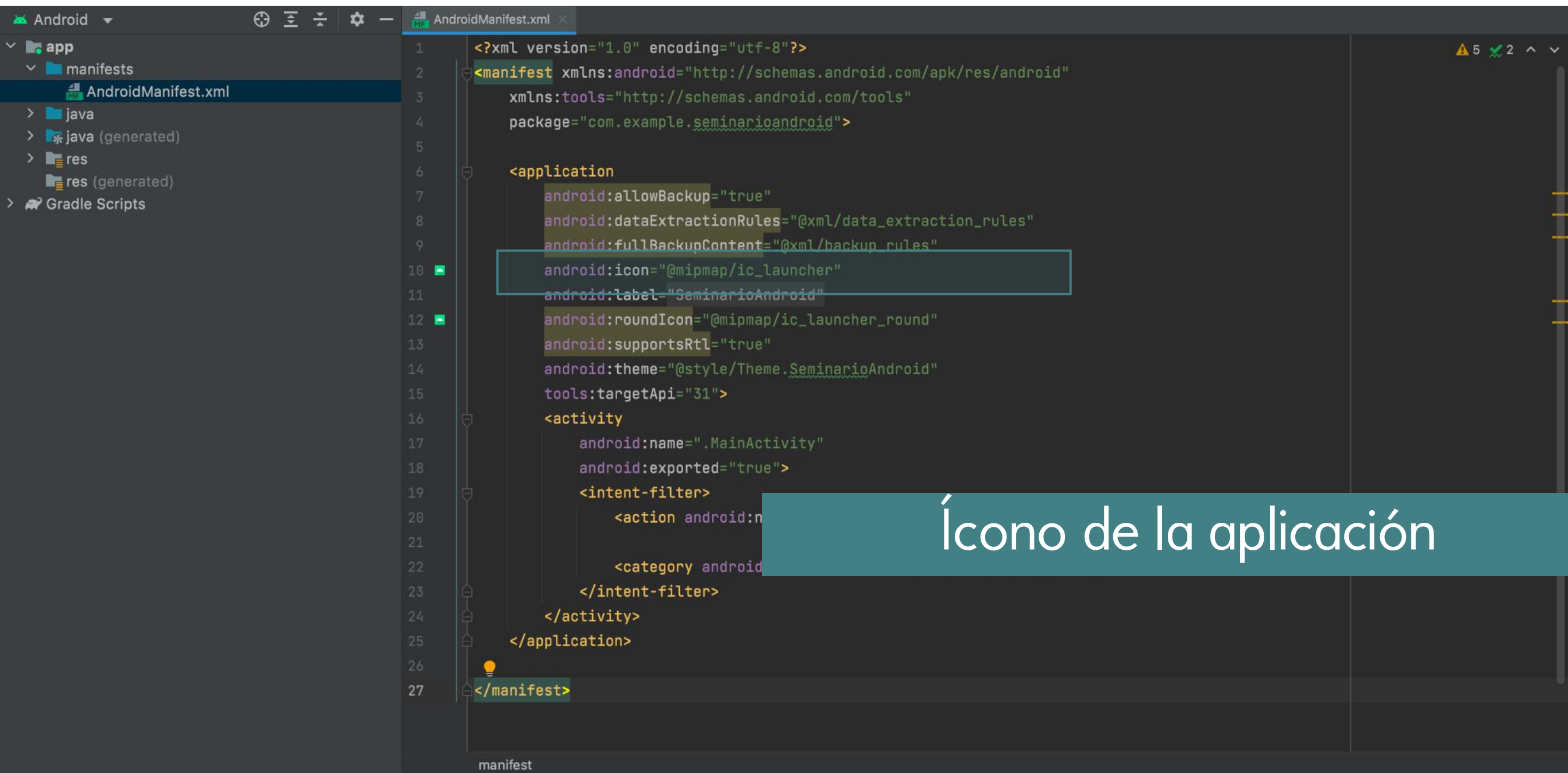
# AndroidManifest.xml



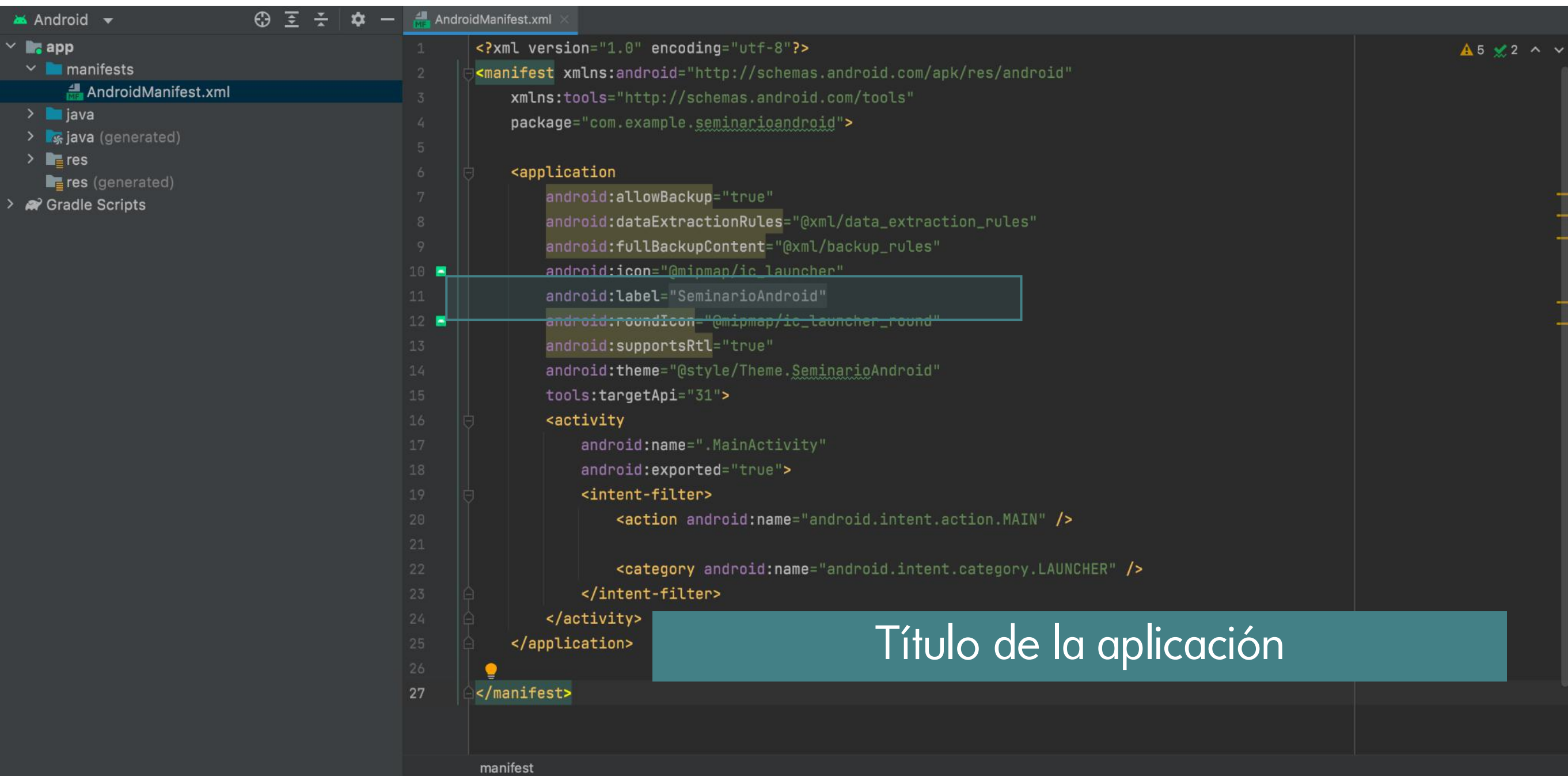
```
1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3    xmlns:tools="http://schemas.android.com/tools"
4    package="com.example.seminarioandroid">
5
6    <application
7        android:allowBackup="true"
8        android:dataExtractionRules="@xml/data_extraction_rules"
9        android:fullBackupContent="@xml/backup_rules"
10       android:icon="@mipmap/ic_launcher"
11       android:label="SeminarioAndroid"
12       android:roundIcon="@mipmap/ic_launcher_round"
13       android:supportsRtl="true"
14       android:theme="@style/Theme.SeminarioAndroid"
15       tools:targetApi="31">
16
17       <activity
18           android:name=".MainActivity"
19           android:exported="true">
20
21           <intent-filter>
22               <action android:name="android.intent.action.MAIN" />
23               <category android:name="android.intent.category.LAUNCHER" />
24           </intent-filter>
25       </activity>
26   </application>
27 </manifest>
```

Cuando esta propiedad vale *true* se hace un respaldo de los datos de la aplicación en la cuenta de GoogleDrive del usuario

# AndroidManifest.xml



# AndroidManifest.xml



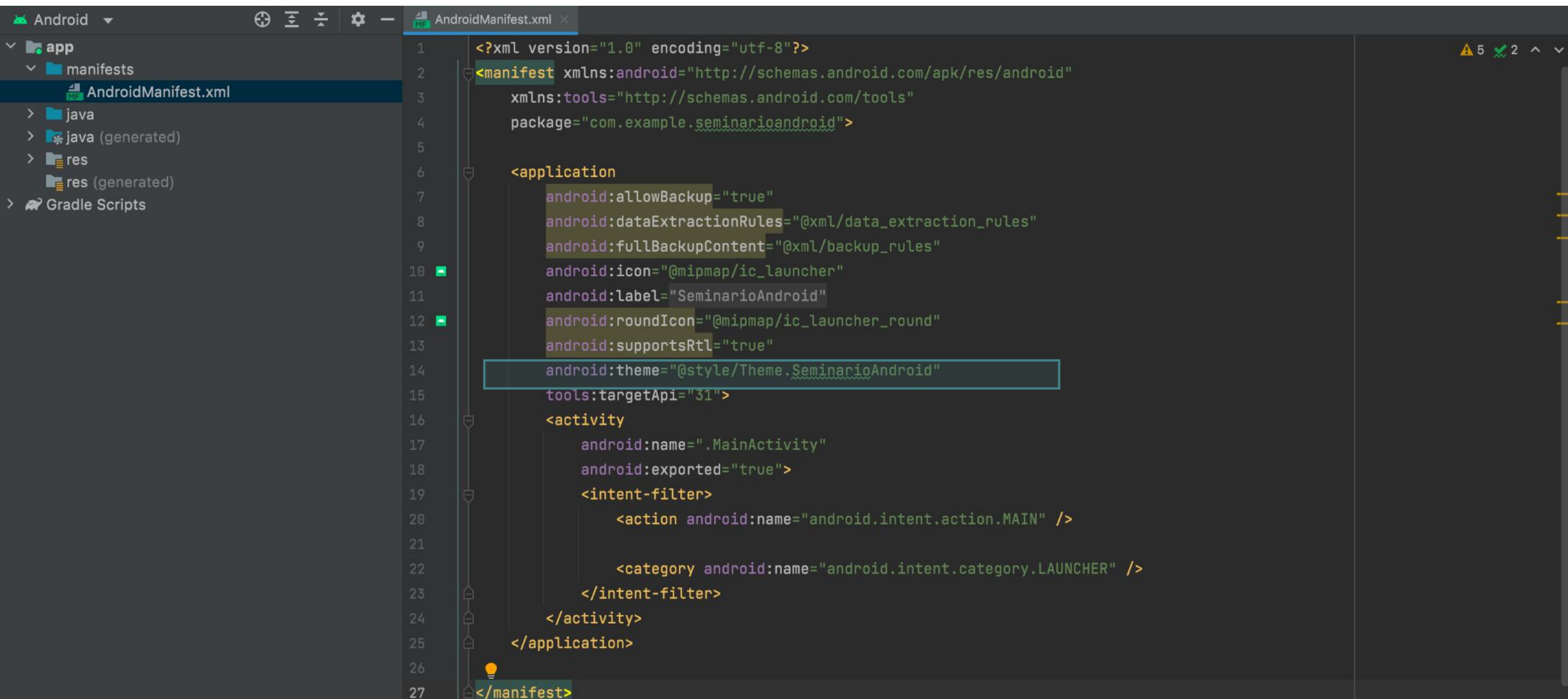
# AndroidManifest.xml



Si la opción está habilitada, en caso de que el usuario configure el dispositivo con un idioma de lectura de derecha a izquierda (Right To Left) la interfaz se acomodará automáticamente



# AndroidManifest.xml



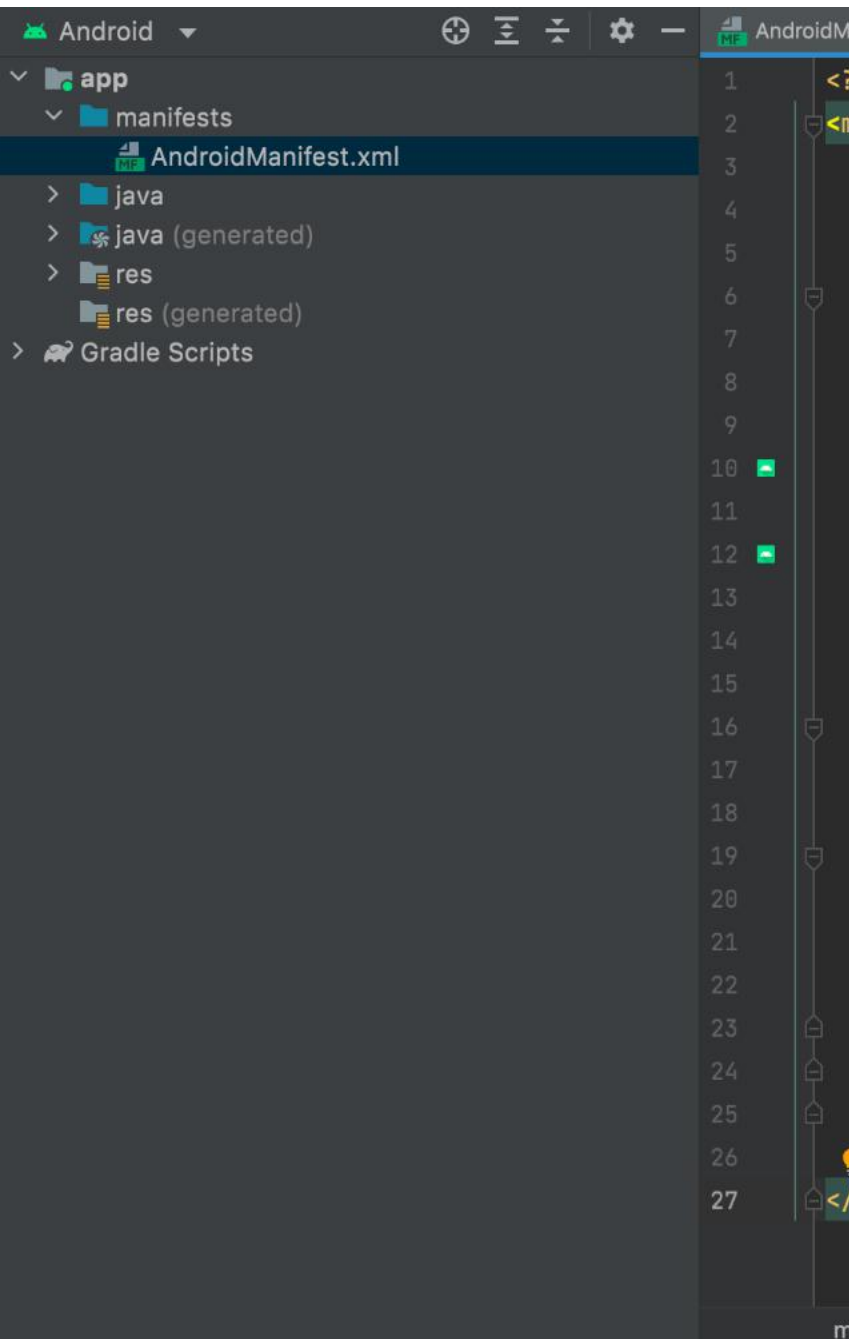
```
1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools"
4          package="com.example.seminarioandroid">
5
6      <application
7          android:allowBackup="true"
8          android:dataExtractionRules="@xml/data_extraction_rules"
9          android:fullBackupContent="@xml/backup_rules"
10         android:icon="@mipmap/ic_launcher"
11         android:label="SeminarioAndroid"
12         android:roundIcon="@mipmap/ic_launcher_round"
13         android:supportsRtl="true"
14         android:theme="@style/Theme.SeminarioAndroid"
15         tools:targetApi="31">
16
17         <activity
18             android:name=".MainActivity"
19             android:exported="true">
20             <intent-filter>
21                 <action android:name="android.intent.action.MAIN" />
22
23                 <category android:name="android.intent.category.LAUNCHER" />
24             </intent-filter>
25         </activity>
26     </application>
27 </manifest>
```

Tema de estilos de la aplicación.  
*En clases posteriores estudiaremos este tema*

# AndroidManifest.xml

Además, describe cómo está compuesta la aplicación.

*Más adelante estudiaremos que existen 4 tipos de “componentes de aplicación” posibles (Activities, Services, Broadcast receivers, Content Providers)*



# AndroidManifest.xml

My Application – AndroidManifest.xml [My\_Application.app]

Refactor Build Run Tools VCS Window Help

androidManifest.xml

AndroidManifest.xml x

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3     package="com.example.myapplication">
4
5     <application
6         android:allowBackup="true"
7         android:icon="@mipmap/ic_launcher"
8         android:label="@string/app_name"
9         android:roundIcon="@mipmap/ic_launcher_round"
10        android:supportRtl="true"
11        android:theme="@style/Theme.AppCompat"
12    >
13        <activity
14            android:name=".MainActivity"
15            android:exported="true">
16            <intent-filter>
17                <action android:name="android.intent.action.MAIN" />
18                <category android:name="android.intent.category.LAUNCHER" />
19            </intent-filter>
20        </activity>
21    </application>
22
23 </manifest>
```

Información para el S.O. de cómo se invoca a la actividad.

*Lo estudiaremos también próximamente.*

# Permisos de usuario en AndroidManifest.xml

Permisos que el usuario debe aceptar para que la aplicación pueda acceder a un recurso

```
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
<uses-permission android:name="
```

```
android.permission.ACCEPT_HANDOVER
android.permission.ACCESS_BACKGROUND_LOCATION
android.permission.ACCESS_BLOBS_ACROSS_USERS
android.permission.ACCESS_CHECKIN_PROPERTIES
android.permission.ACCESS_COARSE_LOCATION
android.permission.ACCESS_FINE_LOCATION
android.permission.ACCESS_LOCATION_EXTRA_COMMANDS
android.permission.ACCESS_MEDIA_LOCATION
android.permission.ACCESS_MOCK_LOCATION
android.permission.ACCESS_NETWORK_STATE
android.permission.ACCESS_NOTIFICATION_POLICY
android.permission.ACCESS_SURFACE_FLINGER
Press ← to insert, → to replace
```



# Permisos de usuario en AndroidManifest.xml

Gestión de permisos antes de Android 6

Ejemplo de acceso a internet

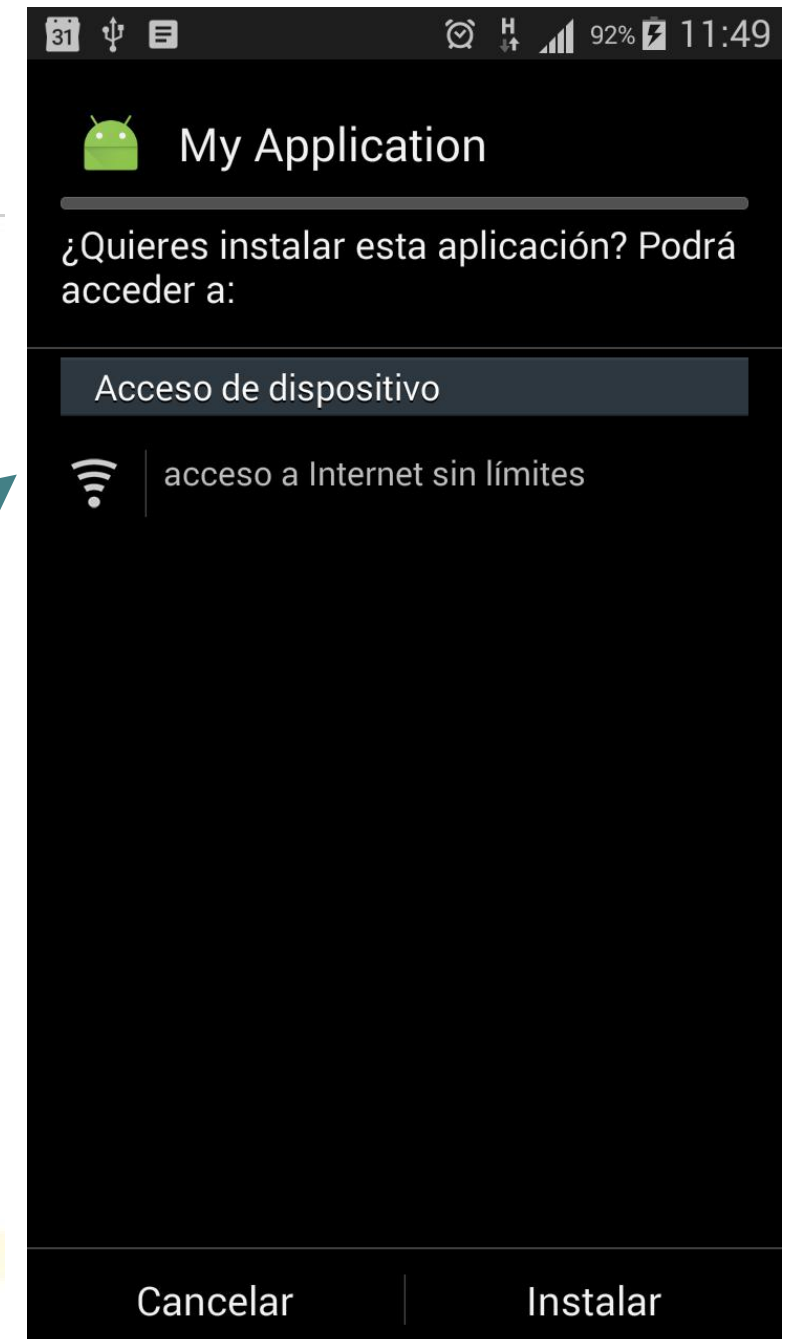
```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.ldelia.myapplication">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/AppTheme">

        <activity
            android:name=".MainActivity"
            android:label="@string/app_name"
            android:launchMode="singleTop"
            android:theme="@style/AppTheme">
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </activity>

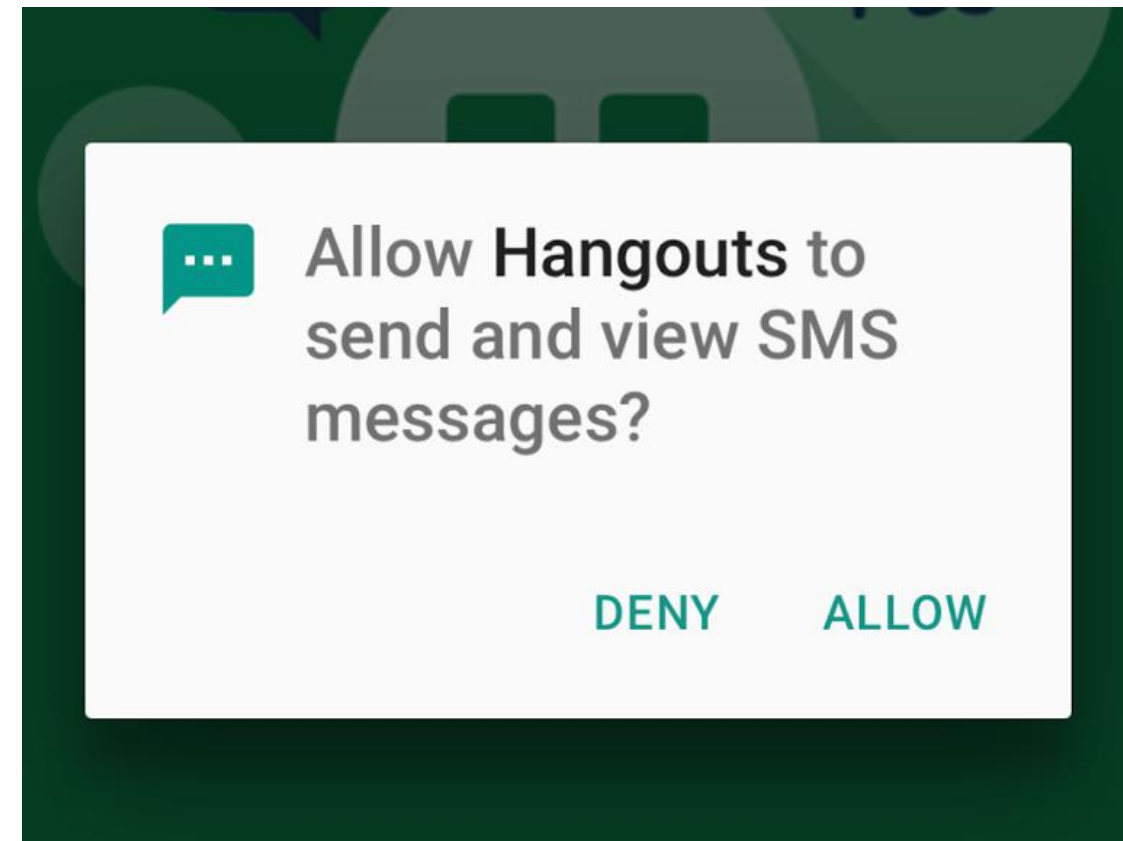
        <uses-permission android:name="android.permission.INTERNET" />

    </application>
</manifest>
```



# Permisos de usuario en AndroidManifest.xml

- En versiones inferiores a Android 6 el usuario acepta los permisos en el momento de la instalación
- Desde Android 6 los usuarios aceptan los permisos cuando la aplicación se está ejecutando.
  - Simplifica el proceso de instalación.
  - El usuario tiene mayor control. Puede aceptar algunos permisos y otros no



Durante el curso se seguirá analizando el contenido del archivo `AndroidManifest`, a medida que se introduzcan nuevos conceptos